

Problem Identification Capstone_Three

Problem Statement Worksheet (Hypothesis Formation)

Which characteristics of contents of an e-book can sufficiently meet a reader?

Build and train a recommendation model in one month for personalized book recommendations based on reader segmentation to improve the accuracy of recommendation to 60%.

1. Context

When readers choose books, they are influenced by different factors such as the type of book, author, publisher, price, whether it is recommended by editors, whether it is a popular book, and other readers' reviews. How can we recommend suitable books to different types of readers?

2. Criteria for success

Build and train a recommendation model for personalized book recommendations to improve the accuracy of recommendation to 60%.

3. Scope of solution space

We hope to use the information in the database to segment readers and form different market groups. At the same time, the types and contents of e-books are also distinguished into different product types.

We hope to use recommendation algorithms to achieve the best match between readers and products.

4. Constraints within solution space

Some readers may seldom conduct product reviews, or the review information for some products may be very limited, which may reduce the accuracy of the recommendation.

5. Stakeholders to provide key insight

- Manager of marketing
- Community of readers
- Retailer of e-books
- Publisher of e-books

6. Key data sources

- [Kaggle.com/ Amazon Kindle Books Dataset](https://www.kaggle.com/datasets/amazon-reviews-51)
Amazon Kindle Books Dataset 2023 (130K Books)